

Joint Score for the Laplace AR(1) Chain under OU Corruption

A working note on Markov structure, innovation coordinates, Fourier representation, and the Hessian field

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Abstract

This note consolidates the working session devoted to Jérôme’s three-step research programme for the Laplace AR(1) toy model under Ornstein–Uhlenbeck corruption. It proceeds in baby steps and ends with an honest assessment of which steps deliver structural insight on the joint score and which deliver only computational tools. The central operational outcome is the identification of the posterior Hessian field $\mathcal{H}_t(\mathbf{x}) = -\nabla_{\mathbf{x}}^2 \log P_t(\mathbf{x})$ as the right object to study going forward.

1 Setup and notation

The clean chain of length K is defined recursively via

$$a_0 \sim \tfrac{1}{2}e^{-|a_0|}, \quad a_k = \alpha a_{k-1} + \varepsilon_k, \quad \varepsilon_k \stackrel{\text{iid}}{\sim} \text{Laplace}(0, 1), \quad |\alpha| < 1. \quad (1)$$

Markovianity makes the joint clean law factorise as

$$P_0(\mathbf{a}) = p_0(a_0) \prod_{k=1}^{K-1} p(a_k | a_{k-1}) = \frac{1}{2^K} \prod_{k=0}^{K-1} \exp(-|a_k - \alpha a_{k-1}|), \quad (2)$$

with the convention $a_{-1} \equiv 0$. The forward (OU) corruption acts independently on each frame:

$$\mathbf{x} = \mu(t) \mathbf{a} + \sigma(t) \boldsymbol{\xi}, \quad \boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, I_K), \quad \mu(t) = e^{-t}, \quad \sigma^2(t) = 1 - e^{-2t}. \quad (3)$$

The likelihood factorises across frames,

$$p(\mathbf{x} | \mathbf{a}) = \prod_{k=0}^{K-1} p(x_k | a_k), \quad p(x_k | a_k) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x_k - \mu a_k)^2}{2\sigma^2}\right). \quad (4)$$

The prior couples adjacent frames; the likelihood does not. This tension is the source of the structure studied below.

2 The Tweedie reduction (joint, not marginal)

Proposition 1 (Joint Tweedie formula). *For any clean law P_0 and isotropic Gaussian corruption with parameters (μ, σ) , the joint score satisfies*

$$S(\mathbf{x}, t) = \nabla_{\mathbf{x}} \log P_t(\mathbf{x}) = -\frac{\mathbf{x} - \mu \mathbb{E}[\mathbf{a} | \mathbf{x}]}{\sigma^2}. \quad (5)$$

Proof. Differentiate $p_t(\mathbf{x}) = \int P_0(\mathbf{a}) p(\mathbf{x} | \mathbf{a}) d\mathbf{a}$ under the integral sign. Since $\partial_{x_k} p(\mathbf{x} | \mathbf{a}) = -\sigma^{-2}(x_k - \mu a_k) p(\mathbf{x} | \mathbf{a})$, dividing through by $p_t(\mathbf{x})$ and recognising the posterior expectation yields the claim. $\square$

Remark 1. The identity is exact for all P_0 , all K , and all $t > 0$. It uses only the Gaussian structure of the likelihood. The entire role of P_0 is concentrated in the posterior mean $\mathbb{E}[\mathbf{a} | \mathbf{x}] \in \mathbb{R}^K$. The non-Gaussianity of P_0 enters the score only through the nonlinearity of this posterior mean as a function of $\mathbf{x}$.

3 Question 1 — Score via the Markov structure

3.1 Forward–backward decomposition

For any node $k \in \{0, \dots, K-1\}$, the marginal posterior obtained by integrating out all other coordinates factorises as

$$p(a_k | \mathbf{x}) \propto p(x_k | a_k) \cdot b_k(a_k) \cdot m_k(a_k), \quad (6)$$

with the messages

$$m_k(a_k) = \int p(a_{k+1} | a_k) p(x_{k+1} | a_{k+1}) m_{k+1}(a_{k+1}) da_{k+1}, \quad (7)$$

$$b_k(a_k) = \int p_0(a_{k-1}) p(x_{k-1} | a_{k-1}) p(a_k | a_{k-1}) b_{k-1}(a_{k-1}) da_{k-1}. \quad (8)$$

This is the standard belief-propagation / forward-backward recursion. Markovianity of P_0 together with the factored Gaussian likelihood is precisely what allows the global integral to split into two one-dimensional recursions.

3.2 Forward message in closed form

The change of variables $r = a_{k+1} - \alpha a_k$ (innovation) yields

$$\boxed{m_k(a_k) = p_t^{(1)}(x_{k+1} - \mu \alpha a_k),} \quad (9)$$

where $p_t^{(1)}$ is the noised marginal density of a single Laplace variable. The forward message is the universal $K = 1$ density evaluated at a shifted argument; the shift $\mu \alpha a_k$ is exactly what frame k predicts for frame $k + 1$ under the AR(1) dynamics.

3.3 Backward message — causal asymmetry

The backward message $b_k(a_k)$ admits no comparably clean form. Two absolute values — $|a_{k-1}|$ at $a_{k-1} = 0$ and $|a_k - \alpha a_{k-1}|$ at $a_{k-1} = a_k / \alpha$ — partition the integration domain into three regions in which the integrand becomes a half-line Gaussian. The backward message is therefore a sum of error functions.

Observation 1 (Causal asymmetry). The forward message is clean because integration over a_{k+1} at fixed a_k follows the causal direction of the chain. The backward message is harder because it inverts the transition $a_{k-1} \rightarrow a_k$. In the Gaussian benchmark this asymmetry disappears algebraically; in the non-Gaussian case it is fundamental.

3.4 What the Markov structure buys

A naïve evaluation of $\mathbb{E}[a_k \mid \mathbf{x}]$ requires a K -dimensional integral. Markovianity reduces this to K sequential one-dimensional integrals via the message recursion. This is *computational locality*: each node communicates with the rest of the chain through two summary functions. It is *not* statistical locality: the messages themselves carry global information, since long-range correlations are built up recursively.

Remark 2 (On the geometry of dependence). For an interior node k of a long chain, the score $S_k(\mathbf{x}, t)$ depends on the full observation $\mathbf{x}$. The dependence is mediated through exactly two channels — b_k and m_k — but those channels carry information from all other frames. The score is therefore not local in $\mathbf{x}$.

4 Question 2 — Innovation coordinates

Define the linear change of variables $r_0 = a_0$, $r_k = a_k - \alpha a_{k-1}$ for $k \geq 1$. Equivalently $\mathbf{a} = L\mathbf{r}$ with L unit lower-triangular,

$$L = \begin{pmatrix} 1 & & & \\ \alpha & 1 & & \\ \alpha^2 & \alpha & 1 & \\ \vdots & & \ddots & \ddots \end{pmatrix}, \quad \det L = 1. \quad (10)$$

Proposition 2 (Innovation factorisation). *The clean law factorises completely in innovation coordinates,*

$$P_0(\mathbf{r}) = \frac{1}{2^K} \prod_{k=0}^{K-1} e^{-|r_k|}. \quad (11)$$

The r_k are independent standard Laplace variables.

The full AR(1) structure has been absorbed into the change of coordinates. In return, the likelihood ceases to be diagonal:

$$p(\mathbf{x} \mid \mathbf{r}) \propto \exp\left(-\frac{1}{2\sigma^2} \|\mathbf{x} - \mu L\mathbf{r}\|^2\right) = \exp\left(-\frac{\mu^2}{2\sigma^2} \mathbf{r}^\top L^\top L \mathbf{r} + \frac{\mu}{\sigma^2} \mathbf{x}^\top L \mathbf{r} + \text{const}\right). \quad (12)$$

The matrix $L^\top L$ is dense. The AR(1) coupling has moved from the prior into the geometry of the likelihood — it cannot be made trivial in both at once.

Remark 3. This duality is the precise non-Gaussian counterpart of the Gaussian observation that the precision matrix is tridiagonal in original coordinates and *diagonal* in innovation coordinates. In the Gaussian case the cross terms in $\mathbf{r}^\top L^\top L \mathbf{r}$ are absorbed by completing the square against the Gaussian prior. In the Laplace case there is no such absorption.

5 Question 3 — Fourier representation

5.1 The K=1 noised marginal

The noised marginal of a single Laplace variable is the density of $y = \mu r + \sigma \xi$ with $r \sim \text{Laplace}(0, 1)$ and $\xi \sim \mathcal{N}(0, 1)$ independent. Its characteristic function factorises into the two known building blocks,

$$\hat{p}_t^{(1)}(k) = \underbrace{\frac{1}{1 + \mu^2 k^2}}_{\text{Laplace}} \cdot \underbrace{e^{-\sigma^2 k^2/2}}_{\text{Gaussian}}. \quad (13)$$

5.2 The screened Poisson equation

Multiplying through by $(1 + \mu^2 k^2)$ and inverse-Fourier transforming using $k^2 \hat{p} \leftrightarrow -\partial_y^2 p$ yields

$$(1 - \mu^2 \partial_y^2) p_t^{(1)}(y) = G_\sigma(y), \quad G_\sigma(y) = \frac{1}{\sqrt{2\pi}\sigma} e^{-y^2/2\sigma^2}. \quad (14)$$

This is a screened Poisson equation. Its Green's function $G_{\text{ODE}}(z) = \frac{1}{2\mu} e^{-|z|/\mu}$ is itself an exponential kernel of width μ , so the solution is a Gaussian convolved with a Laplace kernel. The screening length $\mu = e^{-t}$ measures how much of the Laplace structure of p_0 survives at noise level t .

5.3 Honest assessment

The Fourier and ODE representations turn convolutions into pointwise products and provide a tractable analytical handle on $p_t^{(1)}$ and the message recursion. They do not, by themselves, deliver a structural insight on Markovianity of the joint score. They are computational tools. The structural content lies in the message-passing decomposition (Question 1) and the causal asymmetry it makes manifest.

6 The Hessian field — the right object

6.1 From posterior covariance to Hessian

Theorem 1 (Stein-type identity). *For the OU corruption likelihood,*

$$\frac{\partial \mathbb{E}[a_k \mid \mathbf{x}]}{\partial x_j} = \frac{\mu}{\sigma^2} \text{Cov}(a_k, a_j \mid \mathbf{x}). \quad (15)$$

Proof. Write $\mathbb{E}[a_k \mid \mathbf{x}] = N(\mathbf{x})/D(\mathbf{x})$ and apply the quotient rule. Each term reduces to a posterior expectation by multiplying and dividing by $p(\mathbf{x} \mid \mathbf{a})$ inside the integrand. Using $\partial_{x_j} \log p(\mathbf{x} \mid \mathbf{a}) = (\mu a_j - x_j)/\sigma^2$ and noting that x_j is constant under the posterior, the cross terms cancel and the residue is the stated covariance. $\square$

Proposition 3 (Hessian of the noised log-density).

$$\mathcal{H}_t(\mathbf{x}) = -\nabla_{\mathbf{x}}^2 \log P_t(\mathbf{x}) = \frac{1}{\sigma^2} I - \frac{\mu^2}{\sigma^4} \text{Cov}(\mathbf{a}, \mathbf{a} \mid \mathbf{x}). \quad (16)$$

6.2 Two limits and the bump in between

Observation 2 (Limits of the off-diagonal coupling). The off-diagonal entry $(\mathcal{H}_t)_{jk}(\mathbf{x}) = -\frac{\mu^2}{\sigma^4} \text{Cov}(a_j, a_k \mid \mathbf{x})$ for $j \neq k$ vanishes in both extremes:

- As $t \rightarrow 0$: the likelihood concentrates and the posterior collapses to a point, so $\text{Cov}(a_j, a_k \mid \mathbf{x}) \rightarrow 0$ pointwise.
- As $t \rightarrow \infty$: the prefactor $\mu^2/\sigma^4 \rightarrow 0$ even though the posterior covariance approaches the prior covariance.

The Markov coupling visible in the score is therefore concentrated at *intermediate* diffusion times.

6.3 Within-quadrant precision in innovation coordinates

Inside any single quadrant of (a_0, r) space (defined by the signs of a_0 and r), the absolute values become linear and the posterior is locally Gaussian with precision matrix

$$\Lambda = \frac{\mu^2}{\sigma^2} \begin{pmatrix} 1 + \alpha^2 & \alpha \\ \alpha & 1 \end{pmatrix}, \quad \det \Lambda = \frac{\mu^4}{\sigma^4}. \quad (17)$$

Transforming back to (a_0, a_1) coordinates via $\Sigma_{(a_0, a_1)}^{\text{within}} = L\Lambda^{-1}L^\top$ yields the surprising result

$$\Sigma_{(a_0, a_1)}^{\text{within}} = \frac{\sigma^2}{\mu^2} I. \quad (18)$$

The within-quadrant covariance is purely isotropic. Any off-diagonal coupling in the full posterior covariance $\text{Cov}(\mathbf{a}, \mathbf{a} \mid \mathbf{x})$ must therefore originate from the *between-quadrant* weights of the four-component mixture, not from the local curvature.

7 Summary table

Object	Gaussian AR(1)	Laplace AR(1)
Score $S(\mathbf{x}, t)$	Linear in $\mathbf{x}$	Nonlinear in $\mathbf{x}$
Posterior $p(\mathbf{a} \mid \mathbf{x})$	Gaussian, $\mathbf{x}$ -independent covariance	Mixture of 2^K truncated Gaussians
Hessian $\mathcal{H}_t(\mathbf{x})$	Constant matrix Σ_t^{-1}	$\mathbf{x}$ -dependent field
Off-diagonal at $\alpha = 0$	Vanishes	Vanishes
Locality at $t = 0$	Tridiagonal precision	Distributional curvature on kink lines

8 What is established and what is open

Established.

1. The joint score reduces, via Tweedie, to the posterior mean of the clean signal.
2. Markovianity of P_0 implies a forward–backward decomposition with computational locality.
3. The forward message is the universal $K = 1$ noised density evaluated at a causally shifted argument.
4. The backward message has no clean closed form; the asymmetry is a genuine consequence of the AR(1) causal direction.
5. Innovation coordinates trivialise the prior at the cost of densifying the likelihood.
6. The natural object encoding the structure of the score field is the posterior Hessian $\mathcal{H}_t(\mathbf{x})$, which equals the identity minus a multiple of the posterior covariance of the clean signal.
7. Within each sign-quadrant in innovation coordinates, the local covariance is isotropic. Off-diagonal coupling in the full Hessian is mixture-driven, not curvature-driven.

Open.

1. Closed-form or quadrature-tractable expression for $\text{Cov}(\mathbf{a}, \mathbf{a} \mid \mathbf{x})$ at $K = 2$, including its dependence on $(\mathbf{x}, t)$.
2. Quantitative analysis of the off-diagonal entry $(\mathcal{H}_t)_{01}(\mathbf{x})$ across the diffusion time $t \in (0, \infty)$, locating the bump that interpolates between the two vanishing limits.
3. Generalisation to $K = 3$, where the interior node receives *both* messages simultaneously and is the first genuinely informative case for studying score locality.
4. A precise statement of what “exploitable structure” could mean: sparsity of $\mathcal{H}_t(\mathbf{x})$ in some basis, low-rank corrections to a tridiagonal core, or a controlled perturbation expansion in α or in the non-Gaussianity strength.